

Eric Gabriel Manabat

FULL-STACK AI ENGINEER

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EDUCATION

Pamantasan ng Lungsod ng Valenzuela | Bachelor of Science in Information Technology 2023 - 2027
Honors: **Consistent Dean's Lister (1.03 Running GWA)** | Valenzuela City, Philippines

PROFESSIONAL EXPERIENCE

FlyRank AI | Full-Stack AI Engineer Intern Jun 2026 - Present | Remote

- Architected and deployed an autonomous FastAPI AI sidecar microservice, increasing retrieval accuracy from **81.8% to 86.4% Top-1** on ground-truth benchmarks by engineering hybrid BM25 + dense vector search with Reciprocal Rank Fusion (RRF k=60) and deterministic blend reranking.
- Eliminated out-of-domain hallucinations to achieve a **100% negative query pass rate** and **90% LLM-as-judge relevance score** by designing a bounded 3-step agentic query loop with cosine similarity gating (<0.50 threshold) and inline numbered citations.
- Reduced time-to-first-token (TTFT) by **~65%** and established real-time observability across production deployments by implementing asynchronous Server-Sent Events (SSE) streaming APIs, Prometheus `/metrics` latency histograms, and Grafana telemetry dashboards.

Nexvision Innovations Inc. | Full Stack Software Engineering Intern - Team Lead Jun 2026 - Present | Marikina (Hybrid)

- Built algorithmic payroll computation engine with interval-partitioned time algorithms to isolate 10 PM - 6 AM night differentials from standard overtime multipliers across 4 enterprise applications.
- Automated statutory multi-tier payroll deductions and 13th-month proration across **500+ employee records** by developing modular TypeScript/Next.js calculation engines with immutable audit logging.
- Accelerated team sprint velocity across **180+ Jira/GitHub issues** as Intern Team Lead, conducting peer code reviews for TypeScript type safety and leading architectural oversight.

FEATURED AI PROJECTS

PLV eLib - Academic Library AI Sidecar | *FastAPI, Laravel, PostgreSQL, SQLite FTS5, Prometheus, Grafana* 2026

- Engineered decoupled dual-engine topology pairing Laravel 11 monolith with external FastAPI AI sidecar utilizing SQLite FTS5 (BM25) and dense vectors fused via RRF (k=60) with blend reranking.
- Automated LLM-as-judge evaluation against 27-case golden set (P@5: 0.45, R@5: 0.72, Top-1: 86.36%, Negative Pass Rate: 100%) with bounded citation grounding and deterministic zero-context refusal.
- Implemented Prometheus metrics and a 6-chart Dockerized Grafana telemetry dashboard to monitor inference latency histograms and live query feedback streams.

AGOS - AI-Guided Overflow Surveillance | *FastAPI, React, YOLOv8, OpenCV, Docker, Raspberry Pi* 2026

- Developed solar-powered edge IoT flood surveillance unit running on Raspberry Pi Zero 2W, fusing YOLOv8 object detection, ultrasonic water depth sensors, and weather APIs.
- Decoupled 15s camera streaming from 3-minute server-side inference, cutting compute overhead by **~92%** across a dataset of 5,000 CVAT-annotated drainage images.
- Engineered composite 0-100 hazard scoring engine triggering 3-tier physical LEDs and queued emergency push/SMS alert delivery with a 30-minute deduplication cooldown.

TECHNICAL SKILLS

AI & Machine Learning: RAG Pipelines, Vector Search (pgvector, Hybrid BM25/Dense RRF), Embeddings, YOLOv8, LLM-as-Judge Evaluation, Prompt Engineering, SSE Streaming, Prometheus & Grafana Observability

Languages: Python, TypeScript, JavaScript, SQL, C#, Java, PHP

Frameworks & Full-Stack: FastAPI, Next.js, React, Node.js, Laravel, Tailwind CSS

Databases & Cloud: PostgreSQL, Redis, Supabase, Docker, AWS, GitHub Actions

HONORS & CERTIFICATIONS

- Learn RAG (Retrieval-Augmented Generation)** | Boot.dev | 2026
- Global Cyber Skills Benchmark (Project Nightfall)** - Ranked 63/589 Global Teams | HackTheBox | May 2026
- Google Cybersecurity Professional Certificate** | Google | Jan 2026
- Mastering REST APIs with FastAPI** | Packt / Coursera | May 2026
- ITLympics 2026 - Silver Medal** | Capture the Flag | Pamantasan ng Lungsod ng Valenzuela | Mar 2026